

Export Tiers for Body-Signal Devices

Why a recognised-event interface cannot support cross-person, cross-vendor, or evidentiary use

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Companion specification: ASE-0.1, *Akasara Signal Export* — <https://github.com/breadMSA/Akasara> (branch `spec/ase-0.1`)

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Disclosure of interest: The author is the sole editor of ASE-0.1, the specification whose central requirement this paper argues for, and has a commercial interest in work built above the layer it describes. The argument is offered to be checked, not deferred to. Where it depends on an empirical claim, the claim is cited; where it depends on a definition, the definition is stated.

Abstract

Consumer body-signal devices — surface-EMG bands, EEG headsets, eye and face trackers, and the neural interfaces now entering consumer development — record a continuous physiological signal and export a discrete recognised event: a gesture, a keystroke, a command, a string of text. This paper argues that the choice of export tier is not a quantitative privacy setting but a qualitative boundary, and that three distinct classes of use are not degraded below that boundary but are **structurally unavailable**: (i) processing two people's signals in a common representation, (ii) adapting to a user or a vocabulary the vendor did not train for, and (iii) re-analysing a recording under an interpretation other than the vendor's own — the operation on which both scientific replication and evidentiary challenge depend.

We define a four-tier model (T0 recognised event, T1 feature/latent vector, T2 filtered continuous stream, T3 raw samples), state the projection argument that separates T0 from the rest, and derive three consequences. First, an event record that does not identify the signal frames it was computed from is unfalsifiable, and we describe a minimal mechanism (`t1_seq`) that makes it falsifiable at negligible cost. Second, in jurisdictions applying a reliability standard to expert evidence, a T0-only record can be neither corroborated nor impeached, which is a problem for the party offering it and for the party resisting it alike. Third — and contrary to a common intuition — the commercial value of *interoperating* between vendors is an increasing function of tier: at T0 the exported artefact is already a universal type (text or a label), so bridging is trivial and therefore worth nothing, while the representations for which bridging would be valuable exist only at T1 and above.

We note that the existing international work in this area (ISO/IEC TS 27571:2026; GB/T 47127—2026) standardises how a recording is *written down* and leaves the tier floor, the export obligation, and cross-user processing terms unaddressed. ASE-0.1 is an attempt to occupy that gap; this paper is its argument, stated independently of the requirements text.

Keywords: brain-computer interface, biosignal, data portability, neural data, interoperability, forensic evidence, standardisation

1. Introduction

A device that records a body signal must decide what to hand to software running outside it. Current consumer practice is uniform: the raw signal stays on the device and the interface emits recognised events. Vendors state the rationale in privacy terms, and the rationale is real — a recognised event discloses far less about the wearer than the signal it was computed from.

The claim of this paper is that the same choice has a second consequence which is rarely stated, is independent of the privacy rationale, and is not a matter of degree.

The usual mental model is that an export tier trades fidelity against privacy on a continuous scale: a coarser export is a lossier version of the same thing. For tiers T1 through T3 that model is correct. For the step from T1 to T0 it is wrong, and the error matters because it makes an entire class of applications look merely difficult when it is in fact closed.

2. The tier model

Every body-signal device, whatever the modality, passes internally through the same stages:

sensor	filtered stream	feature / latent vector	classifier	event
T3	T2	T1	T0	

Tier	What crosses the interface
T0	Discrete recognised events: gestures, keys, commands, text
T1	A fixed-dimension numeric feature or latent vector per analysis window
T2	A continuous per-channel stream after fixed, documented filtering
T3	Raw samples at ADC resolution and native rate

The tiers are not a proprietary taxonomy; they are a description of a pipeline that already exists inside every such device. T1 values are computed in the normal course of operation, whether or not they are exported.

2.1 Text is the T0 of a language interface

Where the recogniser’s output is language rather than a gesture label — a neural or muscular interface that decodes to typed characters or to a sentence — the exported artefact is a string. This is still T0. It is a recogniser output projected onto a fixed vocabulary; the vocabulary happens to be a natural language rather than a vendor’s gesture set.

This case deserves separate statement because it is a *harder* case, in three ways. It looks like full fidelity: the user receives the sentence they intended and experiences no loss, and a regulator reading the export policy sees none. The vocabulary is effectively unbounded, so the objection “your label set is too small” is unavailable. And a natural language is a *public* alphabet whose entire function is to mean the same thing across speakers — so projecting onto it removes between-person structure more completely, not less, than a vendor’s private label set does.

3. The projection argument

A T0 event is not a lossy encoding of the signal. It is the image of the signal under a projection onto the vendor's label set:

$f: S \rightarrow L$, where L is a finite (or lexical) set fixed at ship time.

Two consequences follow immediately, and neither is quantitative.

Information outside the label set is not attenuated; it is absent. Any property of the signal that f does not vary with cannot be recovered from $f(s)$ at any data volume, at any sampling rate, over any observation period. Averaging more events does not help, because the events are not noisy measurements of the missing property — they are not measurements of it at all.

The geometry is destroyed, not coarsened. Operations that depend on *distances and directions* in the signal's representation space — the between-person structure that alignment methods exploit, the between-condition structure that a re-analysis would test — do not survive the projection, because L carries no metric that relates back to S .

This is what separates T0 from T1, T2 and T3. Among the latter three the relationship really is one of fidelity: a T1 vector is a compressed, documented summary of the T2 stream, which is a filtered version of the T3 samples. Each retains a metric structure related to the original. T0 does not.

3.1 What is and is not novel here

The first consequence is an instance of the data processing inequality [12]: no function of $f(s)$ can carry more information about s than $f(s)$ does, so post-processing cannot recover what the projection discarded. Stated at that level of generality the point is textbook, and this paper claims no novelty for it.

Three things are added, and they are where any disagreement should be aimed. First, the identification of *where* the discontinuity sits in a real device pipeline — between the feature vector and the classifier output, not anywhere among the signal tiers — which is an empirical claim about how these devices are built, not a mathematical one. Second, the observation that the destroyed property is specifically **metric structure**, which is what distinguishes the T1 T0 step from ordinary lossy compression: a compressed signal supports approximate distance queries, a label set supports none. Third, the mapping from that structural fact onto three concrete failures — cross-person alignment, out-of-distribution service, and adversarial re-analysis — of which the third has,

to the author's knowledge, not previously been raised in the context of consumer biosignal export.

4. Three operations that close at T0

4.1 Cross-person processing

Aligning two people's signals into a common representation is the operation underlying shared-response modelling and hyperalignment, and it is what any person-to-person interface would need. It is defined over vectors: the methods compute a mapping between two representation spaces. Run over event streams the operation does not degrade — it does not exist, because two label sequences have no space to map between. A recent cross-subject result on invasive recordings [6] reports transfer of a pretrained decoder to a held-out subject via a subject-specific projection into a shared space; the reported effect is modest and the decoding target is a language-model embedding, so it should not be read as establishing how much meaning the neural signal itself carries. What it does illustrate is the shape of the operation: the projection is a map between feature spaces, and there is no T0 analogue of it — not a weaker one, none.

4.2 Out-of-distribution users and out-of-vocabulary intents

A user whose signal the vendor's classifier handles poorly, or an intent the vendor never shipped a label for, cannot be served by anyone else at T0. Not because the third party's model would be weak, but because the third party receives only the classifier's own verdict. The failure is invisible in the export: a misrecognition and a correct recognition are the same object.

4.3 Re-analysis under a different interpretation

This is the operation that scientific replication and adversarial legal examination have in common. Both consist of taking a record and asking whether a different interpretation fits it better. At T0 the record *is* the interpretation. There is nothing left to re-interpret.

5. Making an event falsifiable

If T0 events are exported alongside T1 frames, one cheap mechanism converts an assertion into a checkable claim. Each event carries `t1_seq` : the sequence

number of the last exported T1 frame the recogniser consumed before emitting it.

The mechanism costs a 4-byte field. What it buys is that a recipient can re-derive the event from the frames it names, and can detect a device that emits events not supported by the signal it exported. Without it, a device may ship a feature stream and, beside it, an event stream computed from something else entirely, and no external party can tell. Where the recogniser does not run on exported frames, the correct behaviour is to omit the field rather than attach the nearest sequence number and imply a provenance that does not exist.

A T0-only architecture structurally cannot offer this, because there is no exported evidence for an event to point at.

6. Evidentiary consequence

Data from consumer wearables has already been offered in criminal proceedings. As reported in the sources cited here: in an Australian homicide trial the deceased's smartwatch record was used to fix the time of the attack and to contradict the accused's account [7]; a Fitbit heart-rate record played a comparable role in a Californian case, and a Garmin record in a Georgia case in 2024 [8]. A recent systematic review addresses the general question of whether such a device is a reliable witness [9]. These are cited as evidence that the category is already in use, not as authority on any point of law.

Everything offered so far is a *proxy* measurement — heart rate standing in for fear, movement standing in for activity. A body-signal or neural device would offer something nearer the state itself, and the inferential chain becomes correspondingly more consequential in both directions.

Under a reliability standard for expert evidence — *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993), in US federal practice — the tribunal must assess the chain from the instrument reading to the proposition it is offered to prove [10]. The author is not a lawyer, the analysis below is offered as a structural observation rather than as legal advice, and other jurisdictions approach expert evidence differently; what does not vary between them is that some assessment of reliability must be possible at all. A T0-only record defeats that assessment symmetrically:

- The **proponent** cannot show the chain, because the first link is a proprietary classifier whose output is the whole record.
- The **opponent** cannot attack the chain, because there is no underlying observation against which to test an alternative explanation.

The result is not that T0 evidence is weak. It is that its reliability is not assessable at all, which is a worse position for a legal system than either admitting or excluding it on the merits. A tier floor is, among other things, a precondition for such records ever being properly contestable.

7. Interoperability consequence, and a correction to the obvious intuition

It is tempting to argue that a vendor-neutral device could win users by talking to everyone: if two dominant platforms decline to interoperate, a third device that reads both becomes the only one whose user can reach the whole population.

The argument is correct in form and, at T0, empty in substance — and the reason is worth stating because the intuition runs the other way.

At T0 the exported artefact is text or a label. Text is *already* a universal type. Anything can read it, everything already does, and cross-platform text messaging is a solved problem. Bridging two T0 exports is therefore trivial — and worth nothing, because the incompatibility it resolves does not exist. What does exist at T0 is *platform* lock-in at the application layer, which is a different problem with a different remedy, and one that the platform owners themselves have historically solved when it suited them. The precedent is Matter: begun as Project Connected Home over IP in December 2019 and released as Matter 1.0 on 4 October 2022 under the Connectivity Standards Alliance, it was built by the incumbent platform vendors — Apple, Google, Amazon and Samsung among them — once interoperating at that layer stopped threatening anything they valued [13]. The lesson is not that interoperability never arrives; it is that it arrives, from the incumbents, precisely at the layers where it is cheap to them, and therefore that a third party should not plan to be paid for supplying it there.

The exchanges for which a bridge would be genuinely valuable — a representation of intent or state richer than the words it would be flattened into, shared between two people or two devices — exist only at T1 and above. Which yields a compact statement:

The value of interoperating between vendors is an increasing function of the export tier. At T0 it is trivial and worthless; the case where it is worth money is exactly the case where it is not permitted.

The corollary is that a tier floor is not only a research or evidentiary question. It determines whether a market above the platform layer can contain

anything but tenants.

8. Cost of conformance

The tier floor is cheap in a way that distinguishes it from most standardisation proposals: it requires no hardware change. T1 values already exist inside the pipeline; exporting them adds an output path, sequence numbering, session identifiers and a quality field, all bookkeeping.

Two obligations cost real engineering. A **self-test path**, in which the device runs its exact production T1 transform over a fixed synthetic input and publishes the result, is what makes the declared feature space verifiable by a third party rather than merely asserted; the reference specification budgets days for it. And keeping the **previous feature space obtainable** after a change protects downstream users from silent redefinition; it is satisfiable with a downgrade image rather than two live pipelines.

Three vendor objections are worth answering directly rather than dismissing. *Privacy liability*: feature frames are more disclosing than events, which is why the export must be user-directed, locally reachable, and revocable rather than open. *Competitor cloning*: T1 features do help a competitor train a recogniser, and the honest bound is that this is per-subject rather than population-scale, that labels never leave, so the recipient still pays for ground truth, and that a non-adaptive export mode lets a vendor keep personalisation on the device. *SDK gating*: an export conditioned on an approved-developer programme is not an export, because the user's choice of software is precisely what is being conditioned.

9. Relation to existing standards work

ISO/IEC JTC 1/SC 43 (Brain-computer interfaces, formed March 2022) published **ISO/IEC TS 27571:2026**, *BCI data format for non-invasive brain information collection*, in April 2026 [1][2]: basic data elements, technology-specific metadata, a modular structure and naming conventions for EEG, MEG, fNIRS and fMRI. (The description here is taken from the published scope; the author has not obtained the full text, which is behind a paywall, and any characterisation of its detailed provisions should be checked against it.) China's SAC/TC28/SC43 published **GB/T 47127—2026** (multimodal data format) on the same layer [3].

The legal environment runs on a third track again. European data-access law gives a user the right to obtain data generated by a connected product and to

direct it to a third party [4], which secures a *pipe* but says nothing about what travels through it — a right to receive whatever the vendor chooses to emit is satisfied by a T0 export. Conversely, the AI Act's prohibition on inferring emotions in workplaces and educational institutions [5] restricts who may lawfully receive an inference, without touching the tier question either. Access law, prohibition law and format standards leave the same gap between them: none of them states a fidelity.

These standardise **how a recording is written down**. They do not oblige a device to provide an export path, do not state a fidelity at which it must do so, and do not address whether two consenting people may have their signals processed together. The layers stack rather than conflict: a format standard says how to write the file; a tier floor says who is entitled to obtain one, and containing what.

The claim “nobody is standardising this” would be false. The narrower claim is the accurate one: **the rights layer is unclaimed**.

10. Limitations

Three, stated so that they are not presented as discoveries by a reader.

The tier model is a description of current device architectures, not a theorem. A device that exposed a learned representation which was neither a documented feature space nor a label set would not sit cleanly on it. Nothing in the argument breaks, but the mapping would need restating.

The evidentiary argument is structural and untested. No decided case is known to the author in which a body-signal or neural device record was admitted or excluded on the reasoning given in §6; the cited cases all concern proxy measurements such as heart rate. The argument predicts a difficulty; it does not report one.

The interoperability argument in §7 depends on an unfalsified assumption — that consumer read hardware capable of producing representations richer than language will exist. If it does not, the tier floor retains its research and evidentiary value and loses its commercial one. That assumption is stated here rather than buried, because it is the load-bearing one.

11. Conclusion

The export tier of a body-signal device is usually presented as a privacy setting. It is also, and independently, the decision that determines whether anyone other than the vendor can ever compute anything on the signal — including the wearer, a researcher attempting replication, a court assessing reliability, and a second person the wearer wishes to communicate with.

T1 is the lowest tier at which those operations exist at all. It is available at the cost of an output path and a terms clause. The specification accompanying this paper [11] states the requirement in testable form, with a machine-checkable schema, a reference conformance suite and a royalty-free patent commitment; this paper is offered so that the argument can be cited, examined and disagreed with independently of that document.

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